

Experiential financial education: A field study of my classroom economy in elementary schools[☆]



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A B S T R A C T

My Classroom Economy is a simulated economy where students have the opportunity to engage in financial decisions on a daily basis. It provides financial literacy instruction without requiring significant classroom time, and is relatively simple for teachers to implement. Compared to schools where My Classroom Economy was delayed or never implemented, students showed strong improvements in financial knowledge. Compared to a traditional lecture-based curriculum, students showed similar gains in learning. Based on standardized math test scores, there was no substitution away from core school curriculum after the program began. Experiential financial education appears to be an effective strategy to teach financial literacy, even at lower grade levels. It is also a relatively low-cost approach that does not require extensive teacher preparation.

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1. Introduction

A growing number of public schools in the US require a high school financial education course to graduate (Council for Economic Education, 2018). In prior studies, high school financial education mandates helped young adults to develop responsible financial management behaviors, at least in states where mandates were carefully implemented (Harvey, 2019; Stoddard & Urban, 2018; Urban, Schmeiser, Collins, & Brown, 2018). High school students are an appropriate age group to focus financial education, since older students take on increasing responsibility for their own finances. Younger

students also might be able to develop some level of financial literacy throughout schooling (Schug & Walstad, 1991; Suiter & Meszaros, 2005). The US Federal Financial Literacy and Education Commission recommended teaching financial education to students as early as kindergarten (Financial Literacy & Education Commission, 2016). At least 45 states include financial literacy in state learning standards before high school grade levels (Council for Economic Education, 2018). However, younger students typically have little money to manage. This might tend to limit the efficacy of financial literacy education at lower grade levels. This study shows that financial education interventions aimed at younger ages can improve student

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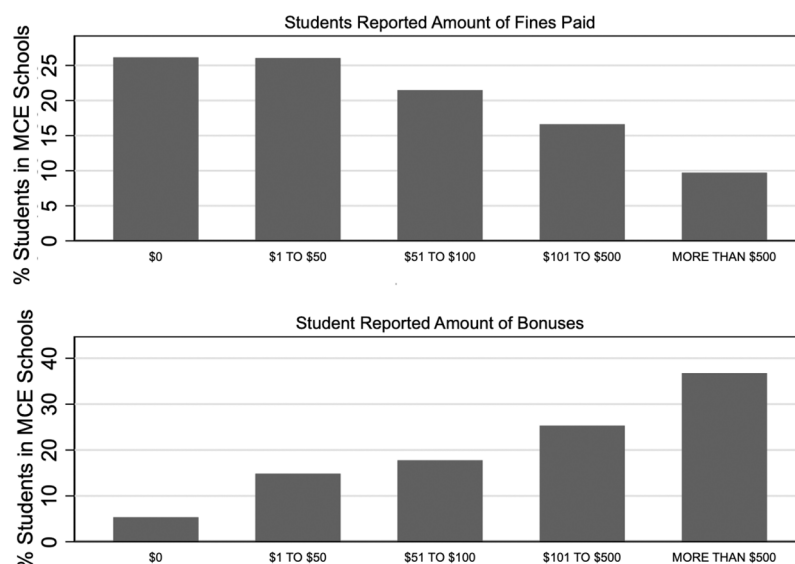


Fig. 1. Student-reported fines and bonuses. Source: Student in-class assessment surveys at follow-up, November 2015. MCE treatment schools only.

financial literacy. This study also illustrates an innovative way of delivering financial education: a simulated classroom economy.

The use of a simulated classroom economy is innovative because it addresses several practical barriers to teaching financial education in elementary schools (Day & Ballard, 1996). One barrier is that many elementary school teachers feel unprepared to deliver personal finance instruction (Way & Holden, 2009). The simulated economy requires less training or preparation than a more formal course, and is less likely to be viewed by teachers as requiring specialized expertise. Teachers can use payments in the economy as a way to reward (or sanction) student behaviors, which has the advantage of serving as a classroom management tool to keep students on task (Rupp, 2014; Waggoner, 2010). Teachers may favor the adoption of such a behavior management strategy even if financial literacy is not their goal. Another barrier to financial literacy courses is that the overall curriculum in most elementary schools is focused on math and language learning standards measured in achievement tests, leaving little time in the curriculum for other topics (Suiter & Meszaros, 2005; Totenhagen et al., 2015). Students in a simulated economy learn through their experiences that are integrated across the school day, relieving the burden on other mandatory components of the curriculum. This approach complements other courses, rather than substituting for other content, and offers even young students a way to practice financial skills and applied economic concepts (Emerson & Taylor, 2004; Harter & Harter, 2009).

My Classroom Economy (MCE) is a program available at no cost to teachers on a website, along with extensive support materials. MCE's learning objectives align with the National Governor's Association Common Core standards for social studies, mathematics and reading, so teachers can achieve multiple learning goals while providing financial education.¹ MCE is designed with varying components based on grade level, including one version that can be used with third through fifth grade classrooms. Core activities include: (1) earning salaries for performing assigned tasks; (2) managing expenses, including paying rent for desks; (3) earning bonuses or incurring fines; and (4) making spending decisions at in-classroom stores or auctions. By being in a classroom with MCE, students naturally engage in experiences that require judging opportunity costs, understanding prices and practicing budgeting, potentially teaching them valuable lessons in financial literacy. Students in MCE may also respond to incentives to engage more

deeply in learning activities overall, boosting academic achievement.

MCE provides standardized materials and procedures that teachers can integrate into their classroom each day. The program requires about three hours of teacher preparation time to set up the economy initially. Teachers can use MCE even if they have no background in social studies or economics, and the rewards and fines in MCE can be used in every class period, and even between class sessions and in non-curricular time.

Studies have examined 'token economies' (see Nelson (2010) for a review), but these studies are on programs focused on curbing negative classroom behaviors and boosting student engagement (see Wardaszko & Jakubowski (2013)). For example, Waggoner (2010) studied a token economy in a high school business class called E-Bucks, finding increased grades and attendance. Other financial education studies have focused on programs like the Stock Market Game (Hinojosa et al., 2007; 2009), or novel teaching tools, such as cartoons (Eissa, Habyarimana, & Jack, 2014; Van Wyk, 2011). To date, no rigorous studies have examined the impact of a simulated economy, including MCE, on elementary school students' financial literacy or overall academic achievement.

The School District of Palm Beach County (SDPBC)—the twelfth largest public school district in the US (Snyder, de Brey, & Dillow, 2016)—implemented MCE in elementary schools starting in the fall of 2015. Principals from 20 schools volunteered to be part of the program. Of these 20 schools, ten were randomly selected to implement MCE in the first trimester, with the remaining ten prevented from implementing MCE until later in the school year. A total of 1805 students in these 20 schools took a financial literacy quiz at the start of the 2015–2016 school year in late August, and then again after the first trimester ended in November 2015. Schools assigned as an MCE 'treatment' group were then compared to the ten schools prevented from operating MCE in the first trimester, serving as a 'control' group. The SDPBC tracked administrative data, including math test scores, for 1078 students, including students in ten schools where MCE was never implemented and the principals never expressed any initial interest in being part of the program. This provides a unique setting to estimate the effects of MCE on the financial literacy of elementary school students. Since MCE promotes student engagement and provides an applied math experience, the program could enhance math learning and therefore math test scores. On the other hand, MCE could crowd out other learning and result in lower standardized math test scores. This study setting allows us to evaluate both the potential effects of MCE on financial literacy and any effects on math test scores.

¹ See <https://www.myclassroomeconomy.org/grades/grade4-5.html> for more detail.

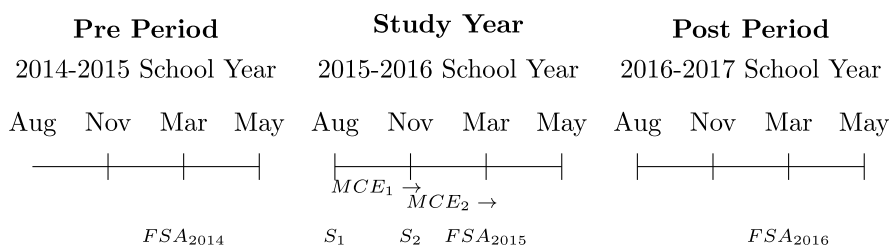


Fig. 2. Overall timeline: three school years (2014-15, 2015-16, 2016-17), MCE pilot and MCE controls, and math testing by trimesters. *Notes:* MCE₂ Control schools began MCE after November 2015. FSA is Florida Standardized assessment for mathematics, conducted annually (called FCAT in 2014). S₁ is student survey conducted in August 2015; S₂ student survey conducted in November 2015. MCE₁ Pilot Schools began MCE in August 2015. No-MCE Schools never offered MCE.

Based on this field study, we find statistically significant improvements in students' financial knowledge from baseline to follow-up at 'treatment' schools where MCE is implemented compared to 'control' schools where principals were interested in using MCE, but did not implement the program. The effect sizes from MCE are similar to the quiz score gains shown in a prior study of a more traditional, classroom lesson plan-based financial education program by Batty, Collins, and Odders-White (2015). Although MCE does not demonstrate a statistically significant improvement in math achievement scores, MCE does not have a negative effect on math achievement scores either. The overall evidence is that MCE is an effective mechanism to teach financial literacy among elementary-school students. This study adds to growing evidence on school-based financial education (see Kaiser and Menkhoff (2019) for a review). The MCE approach may be especially attractive because it does not require extensive teacher support or detract from other curricula.

2. Background

While Palm Beach, Florida, is known as a wealthy enclave, Palm Beach County is diverse. In 2015, 61 percent of SDPBC elementary students qualified for subsidized lunches, and 18 percent were English language learners (Palm Beach County School District, 2016). The SDPBC is a large school district, with more than 100 elementary schools. Prior to the academic year 2015-2016, SDPBC did not formally promote or support MCE in any of the elementary schools in the district.

In the spring of 2015, the SDPBC agreed to test the use of MCE with fourth grade students starting in the fall 2015. The selection of this age group was intentional. These students are around ten years of age, beyond the "five to seven shift" in self-control, planning, and formal decision-making abilities (Doss, Marlowe, & Godwin, 1995; Melton, 2013; Scheinholtz, Holden, & Kalish, 2012; Schug, 1987; Webley & Nyhus, 2006), and are well-suited to experiential learning in MCE. Students at this age can apply financial concepts related to saving money, ownership, planning, and deferred consumption. They are also able to independently complete surveys and financial literacy quizzes, which makes measuring gains in financial knowledge feasible for the study. This age group of students also have the basics of addition, subtraction and multiplication needed to be successful in MCE.

In the 2015-2016 school year, SDPBC assigned a financial literacy coordinator at the elementary-school level to work across the district on the implementation of MCE. The coordinator convened principals in the spring of 2015 to describe the MCE program and to invite principals from traditional, non-charter, elementary schools to take part in the program. A total of 24 school principals, volunteered for the MCE pilot. Four of these schools, with just under 200 students, were dropped from the pilot, leaving 20 schools.²

One complication of a study where schools implement a new education program is that the leadership of some schools may be more willing to use new learning approaches, which also proxies for a general commitment to high-quality education in ways that could bias study

results (Allcott, 2015). This study began with a set of 20 schools where all of the administrative leadership wanted to implement MCE. Ten schools were randomly selected to offer MCE at the beginning of the school year, and ten were not permitted to implement MCE until later in the school year. This approach mitigates the potential selection bias of the most engaged schools also being the schools most likely to implement a novel program.

The district did not compel principals or schools to take part in MCE, but the financial literacy coordinator did monitor activity at schools to make sure teachers were using MCE as intended. Based on data on MCE collected in schools, most students at MCE-assigned schools took part in the program as intended. For example, Figure 1 shows that three-out-of-four students reported paying a fine, consistent with the classroom management aspects of MCE. Nearly 95 percent of students reported receiving a bonus for good behavior or academic achievements. Students also reported a median MCE account balance of 1000 MCE 'dollars', suggesting students were saving over the course of the trimester.

The design of the study includes ten 'MCE treatment' schools where MCE was implemented in the first trimester of the 2015-2016 school year, and ten 'MCE control' schools where principals volunteered to offer MCE in the 2015-2016 school year, but agreed to wait until after the first trimester. In addition, SDPBC provided administrative data for ten additional schools where the principals expressed no interest in MCE and MCE was not implemented at all. This administrative data included standardized math test schools and basic demographics for a three-year panel of students across all 30 schools from the fall of 2014 through the spring of 2017. The following sections describe these data, and the use of student and administrative data to estimate the effects of MCE.

2.1. Data

Figure 2 shows the timeline for the MCE pilot study. The 2014-15 school year was the year before the MCE pilot. During this period, only SDPBC administrative data are available. In the spring of 2015, students completed state standardized math tests.³ The 2015-2016 school year was the year of the MCE pilot. In the first trimester (August to November), ten schools in the MCE 'treatment' cohort implemented MCE. Another ten schools waited to implement MCE until after the first trimester, serving as an MCE 'control' group. The 2016-2017 school year was the post-MCE pilot period. We have no information on which schools used MCE in this period, but we can observe students' test scores based on the school they attended the prior year.

This study has two main parts. First, we examine changes in student financial knowledge based on in-class assessments, conducted only in the ten MCE treatment and ten MCE control schools. Second, we use annual math test score data from students across 30 schools to estimate the effects of MCE on academic achievement using a panel of administrative data. We also compare the financial knowledge of students to

² Two were dropped due to fourth-grade teachers using MCE on their own in prior years; two schools were dropped due to very low student enrollment in grade four.

³ The Florida Standardized Assessment (FSA) in math is administered at elementary schools each year. In the 2014-2015 school year, these were called FCAT (Florida Comprehensive Assessment Test), but transitioned to FSA in the 2015-2016 school year.

Table 1
MCE School cohorts summary statistics.

Variable	MCE Control	MCE Treatment	Diff
White	0.50 (0.02)	0.37 (0.02)	−0.130*** (0.023)
African American	0.12 (0.01)	0.26 (0.02)	0.136*** (0.018)
Latino	0.28 (0.01)	0.27 (0.02)	−0.010 (0.021)
Other	0.10 (0.01)	0.10 (0.01)	0.004 (0.014)
ELL 14	0.13 (0.01)	0.22 (0.01)	0.095*** (0.018)
N	1019	786	1805

Standard errors in parentheses * $p < .1$, ** $p < .05$, *** $p < .01$. Source: SDPBC Administrative Data. MCE treatment schools used MCE in 1st trimester of 2015. MCE control schools did not use MCE in the 1st trimester, but could use after the follow-up survey was conducted. ELL 14 indicates student was English language learner in 2014.

the study by Batty et al. (2015), allowing us to compare the effect sizes of MCE to a traditional lecture-based financial education curriculum.

Table 1 shows the characteristics of students and schools in the MCE sample. About 50 percent of the students in the ten MCE control schools are white, a higher share than the ten MCE treatment schools. The MCE treatment schools have a higher share of African American students, at 26 percent compared to 12 percent in MCE control schools. MCE treatment schools are also dissimilar with respect to English language learner (ELL) status, at nearly twice the rate of the MCE control schools.

Table 2 shows the characteristics of students in the test score administrative data study sample. The sample sizes are smaller than in the MCE pilot study sample, since test score data are only available for a subset of students. The patterns here are quite similar. Panel A compares the MCE treatment and MCE control schools. Panel B compares the MCE treatment to the never-MCE schools. The MCE treatment schools have a higher rate of African American students. MCE treatment schools appear in some aspects to be more disadvantaged than MCE control schools. This could mean students in these schools are ripe for an alternative learning approach, or it could mean these students are unlikely to show positive changes in learning. This may also suggest that estimates that are differences, or changes in outcomes, are preferable to measures of absolute levels.

2.2. Student reports of financial knowledge

Financial literacy among students is measured for 1805 students at 20 schools. At all 20 schools, students completed financial literacy assessment quizzes that were administered in the first week of the trimester, and then again in the last week of the trimester, using the same survey instrument. There are about ten weeks between the baseline and follow-up surveys.

According to SDBPC, 441 students were enrolled in these 20 study schools and did not complete one or more surveys, or completed their surveys but opted out of having their responses used in research, as was required by a human subjects protection-required assent protocol.⁴ It is important to remember that the initial survey and assent to be in the study occurred before students were exposed to MCE, which should limit the extent to which the MCE treatment influenced which students agreed to be included in the study or to complete the baseline assessment. Nonetheless, the estimates we are able to show related to

⁴ The study was reviewed by the school district, the US Treasury, Office of Management and Budget, and a university IRB. Since students were age ten, they were old enough to be required to check a box to assent to have their data used in a research study. Students were told that they would not be penalized if they refused to assent to have their data used for research.

Table 2
Comparing MCE school cohorts and never-MCE schools in test score data.

Variable	MCE Control	MCE Treatment	Diff
Panel A			
White	0.52 (0.03)	0.37 (0.02)	−0.155*** (0.035)
Afr Amer	0.13 (0.02)	0.28 (0.02)	0.146*** (0.029)
Latino	0.24 (0.02)	0.25 (0.02)	0.014 (0.031)
Other	0.11 (0.02)	0.11 (0.01)	−0.005 (0.022)
ELL 14	0.18 (0.02)	0.19 (0.02)	0.007 (0.028)
N	333	454	787
Panel B			
White	0.49 (0.03)	0.37 (0.02)	−0.120*** (0.037)
Afr Amer	0.19 (0.02)	0.28 (0.02)	0.090*** (0.032)
Latino	0.22 (0.02)	0.25 (0.02)	0.028 (0.032)
Other	0.10 (0.02)	0.11 (0.01)	0.003 (0.023)
ELL 14	0.14 (0.02)	0.19 (0.02)	0.046 (0.028)
N	291	454	745

Standard errors in parentheses * $p < .1$, ** $p < .05$, *** $p < .01$. Source: SDPBC Administrative Data. Never-MCE schools did not use MCE at any point. MCE treatment schools used MCE beginning in the 1st trimester of 2015. MCE control schools did not use MCE in the 1st trimester, but could use MCE in the 2nd and 3rd trimesters. ELL 14 indicates student was English language learner in 2014.

financial literacy are based on this subset of students who complied with the data collection process.

Measuring financial literacy among adults is commonly accomplished using a three to six item set of financial quiz questions (for example, see Lusardi, Mitchell, & Curto (2010)). These questions focus on how well people understand compound interest, inflation and investment risk. These types of questions are not appropriate for elementary school students. For this study, we developed a set of financial knowledge quiz questions that would be more appropriate for this population. We began with questions developed by Batty et al. (2015) for a prior study with a similar age group, and then developed additional questions to capture broader dimensions of financial knowledge. These new items were tested with convenience samples of students. The final 13-item multiple choice quiz is shown below, with the correct responses in bold. The quiz includes questions related to basic financial numeracy (questions 1, 2, 4 and 9), financial terms (questions 6 and 11), compound interest (question 10), profit and loss (question 5), account balances (questions 9 and 13), opportunity costs (questions 7 and 12), and the prices of common items (questions 3 and 8). The items below marked with an * are questions Batty et al. (2015) used to study a similar age group.

Financial Literacy Quiz Questions

- David has to pay \$750 in rent for his apartment this month, but he only has \$500 in income. What should he do? (Put \$250 into savings, **Borrow \$250**, Not pay his rent, Don't know)
- David just found a job that pays \$2000 per month. He must pay \$1000 for rent and \$600 for everything else he needs. How long will it take him to save \$800? (1 month, **2 months**, 3 months, 4 months, Don't know)
- Which is closest to what one week's worth of groceries cost for a family of 4? (\$5 \$20 **\$200** \$1000 \$10,000 Don't know)
- Imagine you have to pay \$2 per week to use your desk at school, but you also have the option to buy the desk for \$35 and never pay per week again. If there are 15 weeks left in the school year, is it a

good idea to purchase the desk if you have \$35 you can use to buy your desk today? (Yes, **No**, Don't know)

5. Jane sets up a lemonade stand to sell drinks at the park. She paid \$3 for sugar, \$4 for fresh lemons, and \$3 for cups. Jane made \$12 in revenue from selling lemonade. How much profit did Jane make? (\$1 **\$2** \$3 \$4 \$12 Don't know)
6. People who own things may earn money by renting them to other people. (**True**, False, Don't know)
7. Tracy has \$ 250. She wants to buy a nice backpack for \$100 and buy a new tablet for \$200. She decided to buy a simple \$50 backpack. Tracy must have decided it was more important for her to have a nicer backpack than a tablet. (True, **False**, Don't know)
8. Which is closest to the cost of one ticket to a newly released movie at a regular movie theater? (\$1, **\$10**, \$50, \$75, \$100, Don't know)
9. *The Smiths have \$ 750 in income, and \$ 800 in expenses this month. They are saving money this month. (True, **False**, Don't know)
10. *Suppose you have \$100 in a bank account that pays an interest rate of 10 percent per year. How much would you have in this bank account at the end of 2 years if you leave your account alone? (Exactly \$120, Less than \$120, **More than \$120**, Don't know)
11. *A plan for spending your money is called a: — (**budget**, stock, credit, balance, Don't know)
12. *Ming wanted to buy a fancy notebook for school and save her money to buy a computer. Ming decided to buy a plain notebook that is less expensive so she can save more money for the computer. Ming's decision is an example of — (paying interest, depositing money, **making a tradeoff**, choosing a service, Don't know)
13. *Jill had \$50 in her checking account. She made a withdrawal of \$10 and a deposit of \$20. What is Jill's balance in her checking account? (\$10, \$20, \$50, **\$60**, Don't know)

There are a variety of ways to tabulate quiz scores for study purposes. The classic test scoring method would simply add up the number of correct responses into a summary score, or calculate the ratio of correct responses to the total number of question items recorded. This classical scoring technique places equal weight on every quiz item. In reality, not all quiz questions are equally challenging—some questions are harder and provide more information about the student's underlying knowledge than other questions. For this study, we calculate quiz scores for each student using an item response theory (IRT) model to generate a single-item scale (DeVellis, 2016). The method estimates a parameter for each of the quiz items in terms of how well a correct answer to that question predicts overall performance on the quiz. The IRT approach discriminates between high- and low-performing students, where financial knowledge is the latent trait that the scale is attempting to measure. We used a three-parameter model, including item difficulty, item discrimination, and a guessing parameter, all estimated from the baseline quiz. Scores had a mean of 50 and range from 30 to 70. Changes in student scores from the baseline to follow-up provide a consistent financial literacy test measure within each student.⁵

MCE is not presented to students as a 'financial education class', but rather as a way to gain rewards (and avoid sanctions) in class. However, students in MCE classrooms should be engaged in a wide range of topics that they could perceive as being related to financial education. In order to measure this, students were asked a single survey question item: "how much have your teachers taught you about personal finance?" Students responded using a one-to-five scale, where 1 is 'none' and 5 is 'a lot'. We include this as a subjective measure of what students perceive they are being taught in the classroom. Students are responding about their teacher overall, not specifically MCE. Teachers who are

using MCE may take on more financial literacy topics, or talk more about personal finance in general, and as a result students perceive being taught financial literacy. This measure allows us to test if students recognize that MCE, while only a simulated economy, is directly teaching them personal finance skills. This is a secondary outcome, but these estimates help illustrate how MCE is viewed by students and offer corroborating evidence of MCE as a mechanism to teach financial literacy.

Table 3 shows summary statistics for 1805 students with financial knowledge scores. The MCE treatment and control schools have students with similar scores at the baseline, but the MCE treatment school students show higher scores at the follow-up quiz in November. Students appear to report being taught financial literacy in both sets of school in both periods, but the MCE treatment schools show students with higher rates of reporting being taught financial topics in general. This may be a baseline difference, or students may have been aware of MCE at the start of the new school year and responded accordingly.

2.3. Math achievement test scores

While the primary focus of this study is on how MCE influences student financial literacy, it is also important to consider how the program might impact student learning overall, beyond simply in the domain of personal finance. Using economic incentives, students in MCE classrooms are perhaps more likely to stay on task in their lessons, cooperate more fully in assignments and activities and engage in deeper learning. MCE is designed to address Common Core learning standards, including for mathematics, which is a major focus of later grades in elementary school. It is possible that students in schools using MCE will show greater improvements in terms of standardized math tests than students in other schools without MCE. An alternative is that teachers using MCE end up substituting valuable learning time with MCE activities, and get bogged down in managing the simulated economy. In this context, MCE could improve financial learning at the expense of traditional classroom education across subjects. We attempt to examine the direction of the effects of MCE on broader student achievement using school standardized test scores.

The SDPBC provided standardized math achievement test scores for students. Annual student standardized math achievement test score data are available for 1078 students across the 30 schools. Some students are not observed due to administrative limitations, students moving in or out of the 30 focal schools in the study, not completing a test, and other issues.

The standardized math test is reported on a five-point scale. The lowest score is one, and indicates a student highly likely to need substantial support for the next grade. Five indicates mastery, meaning a student who is highly likely to excel in the next grade. The average school in our data has a mean math test score of about 3.3—at or above state median.⁶ (Figure 4 shows the trends in math score by year across the students in the 30 schools in the study.)

Table 4 shows the summary statistics for math scores. Panel A compares the MCE treatment and MCE control schools. Panel B compares the MCE treatment to the never-MCE schools. The MCE control schools have higher test scores on average in all periods, and the MCE treatment schools have lower scores, again consistent with MCE treatment schools being more likely to be disadvantaged. The focal metric is whether a student's score is increased (Up) for a given pair of test years. For example, Up AY16 (vs 2014) indicates the average rate of students with higher test scores in 2016 than they did in 2014. Students in MCE treatment schools, who had the longest exposure to MCE, were more likely to have increasing math test scores. The MCE control schools implemented MCE for two or fewer trimesters of the 2015-2016 school

⁵ See Knoll and Houts (2012) for a discussion and application to financial education.

⁶ For detail on the Florida Standards Assessments (FSA) test, see <http://www.fl DOE.org/accountability/assessments/k-12-student-assessment/fsa.shtml>

Table 3

MCE School cohorts financial literacy quiz and student reports of being taught personal finance.

	MCE Control	MCE Treatment	Diff
Fin Quiz Score Aug 15	49.79 (0.25)	50.28 (0.28)	0.492 (0.376)
Fin Quiz Score Nov 15	49.33 (0.25)	50.87 (0.30)	1.543*** (0.388)
Taught Personal Finance Aug 15	2.96 (0.04)	3.29 (0.05)	0.328*** (0.063)
Taught Personal Finance Nov 15	2.90 (0.04)	3.57 (0.04)	0.666*** (0.062)
N	1019	786	1805

Standard errors in parentheses * $p < .1$, ** $p < .05$, *** $p < .01$. Source: Student In-class Assessment Surveys. MCE treatment schools used MCE in 1st trimester of 2015. MCE control schools did not use MCE in the 1st trimester, but could use MCE after the follow-up survey was conducted. Quiz Score based on IRT financial literacy quiz. Taught Personal Finance based on student self report on 1 to 5 rating, where 5 is “My teacher provides a lot of personal finance teaching.”

Table 4

Math scores for MCE school cohorts.

Variable	MCE Control	MCE Treatment	Diff
Panel A			
Score Levels (1-5):			
Math 2014	3.37 (0.07)	3.10 (0.06)	−0.263*** (0.091)
Math 2015	3.62 (0.07)	3.22 (0.06)	−0.404*** (0.095)
Math 2016	3.62 (0.07)	3.44 (0.06)	−0.185** (0.092)
Score Increased (0-1):			
Up AY16 (vs 14)	0.36 (0.03)	0.41 (0.02)	0.046 (0.035)
Up AY15 (vs 14)	0.38 (0.03)	0.31 (0.02)	−0.072** (0.034)
Panel B			
Score Levels (1-5):			
Math 2014	3.23 (0.07)	3.10 (0.06)	−0.127 (0.093)
Math 2015	3.41 (0.07)	3.22 (0.06)	−0.185* (0.099)
Math 2016	3.44 (0.07)	3.44 (0.06)	−0.007 (0.096)
Score Increased (0-1):			
Up AY16 (vs 14)	0.38 (0.03)	0.41 (0.02)	0.032 (0.037)
Up AY15 (vs 14)	0.33 (0.03)	0.31 (0.02)	−0.017 (0.035)

Standard errors in parentheses * $p < .1$, ** $p < .05$, *** $p < .01$. Source: SDPBC Administrative Data. MCE treatment Schools used MCE in 1st trimester of 2015. MCE control schools did not use MCE in the 1st trimester, but could use MCE in the 2nd and 3rd trimesters. Never-MCE schools did not use MCE at any point. Florida standardized FSA/FCAT math scores: 1=Inadequate (highly likely to need substantial support for the next grade). 2=Below Satisfactory (likely to need substantial support for the next grade). 3=Satisfactory (may need additional support for the next grade). 4=Proficient (likely to excel in the next grade). 5=Mastery (highly likely to excel in the next grade). Up AY16 (vs 14) indicates score is higher in 2016 than 2014 for that student. Up AY15 (vs 14) is an indicator of a score increased in 2015 compared to 2014.

year. Students in these schools show lower rates of improving test scores than students in MCE treatment schools. The non-MCE schools did not implement MCE in classrooms at any time, and also show lower relative rates of students having higher scores than the comparison testing year relative to students in MCE treatment schools.

2.4. Power and minimum detectable effects

For this study we have one sample of student survey data for estimating the effects of MCE on student knowledge, and then a sub-sample for estimating the effects of MCE on math test scores. The student survey dataset has 1805 students across 20 schools. We can estimate the minimum detectable change in quiz scores, given a 90 percent statistical significance level, an 80 percent power level, and the mean and standard deviation shown in Table 3. Based on these parameters, the 1805 student sample is large enough to be able to estimate a difference in scores for MCE schools of about 0.9 (relative to a mean of about 50). This is about one-eighth of a standard deviation—a relatively small minimum effect size (Hill, Bloom, Black, & Lipsey, 2008a). The math test score sample is smaller. Comparing the MCE treatment school students to MCE control students yields 787 students across 20 schools. Based on the same statistical significance and power assumptions used in the student survey estimates, this sample size will allow us to detect a change of about ten percentage points in the rate of students experiencing an increase in math scores (relative to a mean of 39 percent). The effect size comparing students in MCE treatment schools and never-MCE schools is similar. Combining all three sub-groups across all 30 schools (MCE treatment schools, MCE control schools and never-MCE schools) offers more power to detect effects, but this remains a less robust sample than the quiz score data. Unfortunately, we were unable to access administrative data beyond what was provided by the SDPBC.

3. Estimation

3.1. Estimating the effects of MCE on financial literacy

We estimate changes in student-reported responses within each student, i , in the following specification:

$$Y_i = \alpha_0 + \beta_1 MCE_i + \beta_2 Y_{i,t-1} + X_i + \epsilon_i \quad (1)$$

where the β_1 estimate is the MCE effect on financial literacy for student i , and MCE_i is an indicator of student i being at an MCE treatment school that provided MCE in the first trimester of the 2014–2015 school year. Initially, we estimate β_1 ignoring the baseline level of $Y_{i,t-1}$. Next, we run another β_1 estimate controlling for the baseline measure. This specification also includes X , a set of demographic characteristics including student race and an indicator for a student being designated an English language learner (ELL) the prior year.

The error term, ϵ_i is initially a robust standard error estimate, but then we provide a second estimate with the errors clustered at the school level, which is appropriate in this design where schools are the level of treatment assignment (Cameron & Miller, 2015). Because there are only 20 clusters, we estimate p-values and 90 percent confidence intervals using wild bootstrapped clustering at the school level with 999 repetitions (Cameron, Gelbach, & Miller, 2008).⁷

3.2. Estimating effects of MCE in schools on math achievement test scores

The second identification strategy is similar to that of Eq. (1). The difference in Eq. (2) is that the outcome of interest is an indicator of a student having a score on the state standardized math test that is higher than the comparison year. Variations of this estimate exclude the MCE control schools and only estimate effects for MCE treatment schools and never-MCE schools. A second variation is an estimate of MCE treatment schools and the MCE control schools, excluding the never-MCE schools. The MCE treatment schools provided students a longer exposure to MCE, which could more strongly influence test scores than the MCE control schools, where MCE was implemented for no more than two

⁷ These estimates are executed with the user-written command `boottest` by Roodman (2015).

trimesters in the 2015-2016 school year. The rate of students having higher math test scores is estimated as:

$$Y_i = \alpha_0 + \beta_1 MCE_i + X_i + \epsilon_i \quad (2)$$

where, like in Eq. (1), MCE_i is an indicator of student i being at an MCE treatment school. Controls, X , include student race, English learner status, and the student's 2014-2015 school attendance record. The error term ϵ_i are initially traditional robust error estimates, then wild bootstrapped clustered standard errors at the school level. The outcome, Y_i is defined as 'Up AY16 (vs 14)', which indicates a student's math score was higher in the 2016-2017 school year than it was in the 2014-2015 school year. This measure compares the year prior to MCE, relative to the year after, and is perhaps the strongest estimate of the effects of MCE. Alternatively, Y_i is defined as 'Up AY15 (vs 14)' which is an indicator of a student's math test score for the 2015-2016 school year being higher than in the prior year. Since math tests occur in March, MCE was in operation at most one trimester for the MCE control schools and two trimesters for the MCE treatment schools. The last version of Y_i is 'Up AY16 (vs 15)', which is defined as a math score for that student being higher on the 2016-2017 test than on the 2015-2016 test (Table 6).

4. Results

Table 5 shows estimates based on the specification in Eq. (1), with wild bootstrap-clustered errors at the school level. Panel A is the estimated level of financial literacy scores, and Panel B adds in baseline controls to show changes in scores over the trimester. Both panels show that students at MCE schools have more financial knowledge, and are also more likely to report being taught about personal finances. Students in MCE schools have about 2.2 points higher financial literacy scores at the end of the first trimester than students in the control group schools. Students in MCE schools also show about a 1.5 point increase in scores since the beginning of the trimester. These estimates are relative to a mean of 50, suggesting that students exposed to MCE increased their financial literacy scores by about 3 percent as a marginal effect over the trimester. Moreover, the upper end of the confidence interval would not rule out as much as a 6 percent marginal increase in

Table 5

My classroom economy, estimated effects of MCE treatment vs MCE control cohort, clustering at school level.

	(1) Quiz Score	(2) Taught Financial Literacy
Panel A: Outcome as of November 2015		
MCE Treatment	2.223*	0.630
p	0.10	0.105
90CI	[0.0407, 4.291]	[-0.00989, 1.185]
Controls	Yes	Yes
Baseline Value	No	No
Panel B: Change in Outcome from August to November 2015		
MCE Treatment	1.488**	0.531*
p	0.05	0.09
90CI	[0.211, 2.856]	[0.0123, 0.995]
Controls	Yes	Yes
Baseline Value	Yes	Yes
Mean	50.00	3.193
StDev	8.197	1.344
N	1805	1805
Clusters	20	20

School-clustered standard errors. * $p < .1$, ** $p < .05$, *** $p < .01$. Source: Student In-class Assessment Surveys. Panel A estimates are based on OLS specification without controlling for student's baseline response; Panel B include baseline values. 90% confidence intervals with wild bootstrap clustered errors at school level with 999 repetitions (Cameron et al., 2008)). Column (1) based on IRT financial literacy quiz score. Column (2) student self report on 1-5 rating, where 5 is "My teacher provides a lot of personal finance teaching." Controls include student race and ELL status.

Table 6

Improved math test scores.

	(1) Up AY16 (vs 14)	(2) Up AY15 (vs 14)
Panel A: MCE Cohorts vs Never MCE		
MCE Treatment	0.043	-0.002
p	0.55	0.98
90CI	[-0.0824, 0.163]	[-0.101, 0.0920]
MCE Control	-0.026	0.035
p	0.70	0.56
90CI	[-0.137, 0.0934]	[-0.0620, 0.143]
Clusters	30	30
Mean	0.387	0.336
StDev	0.487	0.473
N	1078	1078
Panel B: MCE 1st Trimester vs. Never MCE		
MCE Treatment	0.0372	-0.0027
p	0.56	0.96
90CI	[-0.0805, 0.155]	[-0.0940, 0.0973]
Clusters	20	20
Mean	0.397	0.317
StDev	0.490	0.466
N	745	745

Note: Wild bootstrap clustered errors at the school level. * $p < .1$, ** $p < .05$, *** $p < .01$. Source: SDPBC Administrative data on Florida standardized math tests. Up AY16 (vs 14) indicates math score is higher than 2014 for that student. Up AY15 (vs 14) is an indicator of a student math score that increased over 2014. MCE treatment schools used MCE in 1st trimester of 2015. MCE control schools did not use MCE in the 1st trimester, but could use MCE after the follow-up survey was conducted. Never-MCE schools never used MCE in any period. Controls include ELL student, student retained, and student race.

financial literacy scores over the trimester due to exposure to MCE. Even the lower bound of these estimates is still positive. The overall effect size for knowledge gains among students in MCE is about one-quarter of the baseline score standard deviation, magnitude that is moderate in size for educational interventions (Hill, Bloom, Black, & Lipsey, 2008b). Considering MCE is not a formal course, and does not take up much of the school day, these effect sizes are notable. It is also reassuring that column (2) shows that students report being taught financial literacy in school at about a half-point higher on a five-point scale, which students see MCE as a learning experience.⁸ Overall, these estimates are consistent with increases in financial literacy quiz scores among students at schools where MCE was implemented compared to students at schools where MCE was not implemented. Over just 10 weeks, students scored higher on a carefully-designed objective quiz.

While Table 5 is promising, we also want to rule out that gains in financial literacy are not coming at the expense of core academic learning. Table 6 shows the estimated MCE effects from Eq. (2) on state FSA math achievement test scores, based on an indicator of a positive change in scores between two years. Panel A compares students in both MCE treatment and MCE control schools to never MCE schools. Panel B shows the MCE treatment school students compared to never-MCE schools, excluding MCE control schools. State math achievement tests are conducted each spring. Students in MCE treatment schools in the fall of 2015 had a longer exposure to MCE before the math test, which could more strongly influence test scores than the MCE control schools, where MCE was implemented for a trimester or less by the spring of 2016.

There are two time periods where we can examine changes in math test score. Column (1) shows the rate of increasing math test scores from the 2014-2015 test to the 2016-2017 test. This is a comparison of

⁸ Although SDPBC records are incomplete due to teachers moving between classrooms and schools, we are able to estimate models with teacher fixed effects for a sub-sample, as shown in Appendix Table 8. The estimates are smaller, but still positive.

the pre-MCE year test to the post-MCE year test students took in the spring of 2017. Column (2) is an estimate of a positive score change from that same spring 2015 test to the spring 2016 test—in this case some students are still in MCE while taking the math achievement test. Both estimates are potentially informative. The former is the effect in the year after MCE was piloted, showing any gains or losses from student engagement in MCE. The latter is the immediate effect of students being actively engaged in MCE on test scores in that same school year.

All the estimates in Table 6 are small in magnitude, and have large standard errors. There is no evidence MCE has a negative association with math test scores. There is a very weak pattern of students in MCE treatment schools experiencing rising scores in the year after MCE compared to the year before, but the estimates for the math test conducted during the year MCE was piloted are close to zero (Appendix Table 9 shows additional estimates).⁹ None of these estimates are close to statistically significant. Table 5 shows students in MCE have increased financial literacy; Table 6 shows that these gains in financial literacy do not come at the cost of lower achievement in math—MCE is not a distraction from core learning objectives.

4.1. Comparing MCE to a standard curriculum

Financial Fitness for Life (FFFL) is a lecture-based financial literacy program offered in elementary schools.¹⁰ There are many similarities between MCE and FFFL. The FFFL curriculum includes teaching materials for 16 lessons, and over the course of the school year requires about 20 hours to fully implement, similar to MCE. Like MCE, FFFL is designed to complement core subjects, rather than serve as a stand-alone financial education class.

Unlike MCE, FFFL is a more formal teaching curriculum, with teacher-led content delivery and student assignments. While FFFL is integrated into core subject classes like math and reading, MCE is operational throughout the school day, including in between class periods or during breaks. The time required to implement the two programs is relatively similar, but the form of the teacher preparation required is different. MCE requires teachers to acquire basic classroom supplies that students can buy at the store or in auctions—the SDPBC implementation of MCE included about \$100 of materials per classroom. Teachers could use donated items, free homework coupons, or a pass for ‘extra free time’ as store items if they did not have these supplies, however. FFFL requires teachers to prepare each of the lesson plans and integrate them into their existing teaching plans. The electronic FFFL teacher guide costs about \$100, including permissions to make copies of student handouts and worksheets as needed.

Unlike FFFL, MCE has the added attribute of potentially helping teachers to improve classroom behavior and student attention. Also, teachers who are reluctant to teach financial education because they not have any background on the subject may want to adopt MCE because it is a classroom management tool. MCE also operates throughout the school day, rather than only as part of designated lesson plans, and may provide students a more enriching experience.

Batty et al. (2015) studied a five-lesson version of FFFL, showing positive effects of the program on student financial literacy. The authors randomly assigned FFFL to 1506 fourth-grade students across 107 classrooms in two school districts—one in Wisconsin and one in Texas. The FFFL program implemented by Batty et al. (2015) included a half-day in-service to train teachers on the curriculum, and required about five hours of formal class time over a six-week study period.

Importantly for this study, Batty et al. (2015) used five of the same financial education quiz questions as used in the MCE pilot. Using the

Table 7

Comparison of standardized financial literacy quiz scores for my classroom economy and financial fitness for life.

	(1) MCE	(2) FFFL
Quiz Score Change	2.053**	2.191***
p	0.05	0.00
90CI	[0.449, 3.627]	[1.529, 2.852]
Mean	50.00	50.05
StDev	6.422	6.401
N	1805	1506
Assignment	School	Classroom
Clusters	20	107

* $p < .1$, ** $p < .05$, *** $p < .01$. Estimates based on common five-item financial literacy quiz. FFFL data described in Batty et al. (2015). OLS estimates of change in score pre-post treatment. 90% confidence intervals with wild bootstrap estimation using clustering at Assignment level with 999 repetitions (Cameron et al., 2008).

same IRT scoring procedures and parameters, we can estimate and compare student knowledge gains across the two studies based on the quiz questions used in both studies. Using the specification in Eq. (1), Table 7 shows estimated improvements in student financial literacy of about 2 points for MCE. This is relative to a mean of 50 and a standard deviation of 6.4—implying an effect size of about 0.31. The FFFL estimates only differ in that random assignment was at the classroom level, rather than school level, resulting in more clusters. For FFFL, using the same scale, the point estimate is about 2.1—for an effect size of 0.34.

The effect sizes are about one-third of a standard deviation in both settings. Using the 90 percent intervals in Table 7, the MCE effect sizes range from 0.07 to 0.56. The FFFL effect sizes range from 0.24 to 0.45. Both programs are effective in terms of improvements in measured financial knowledge of elementary-age students. Figure 3 shows MCE is not a less effective way to teach financial literacy than a more formal lesson-based program. MCE has the advantage of being a classroom management strategy that requires less preparation for teachers to implement.

In a survey of teachers in the MCE treatment schools conducted by the SDPBC, 82 percent of teachers rated their experience with MCE as good or excellent, and 95 percent of teachers reported that they planned to continue to use MCE. It seems likely that teachers can continue MCE for the full school year without much additional preparation; adding more lesson plans using a traditional instructional method in a program like FFFL could be more challenging and incur added costs for preparation and training. MCE is no more costly to implement, and may prove to be more popular with teachers to use with students.

5. Discussion and conclusions

Introducing a new curriculum in schools is challenging. Teachers require training and support, and some teachers may resist changing long-established lesson plans. MCE serves a double duty, as both a financial education program and a classroom management system. Teachers can integrate MCE into the school day without training, which may make teachers more willing to try MCE than a more traditional financial education program.

Students in schools where MCE is introduced appear to ‘learn by doing’. Yet, this study is on the use of MCE for just one trimester. MCE gives students opportunities to receive feedback on their decisions over time, and students may learn more when MCE is used over a full school year. Teachers can also incorporate more elements of personal finance, such as loans, insurance and investments, with more time. A longer exposure to MCE might lead to greater gains by students.

There are other forms of experiential financial learning in schools, such as savings programs or student-run banks or credit unions (Batty

⁹ An estimate for MCE treatment vs. MCE control schools, excluding never MCE, included in the appendix Table 9, also do not show statistically significant patterns.

¹⁰ See <http://fffl.stage.councilforeconed.org/> for more information.

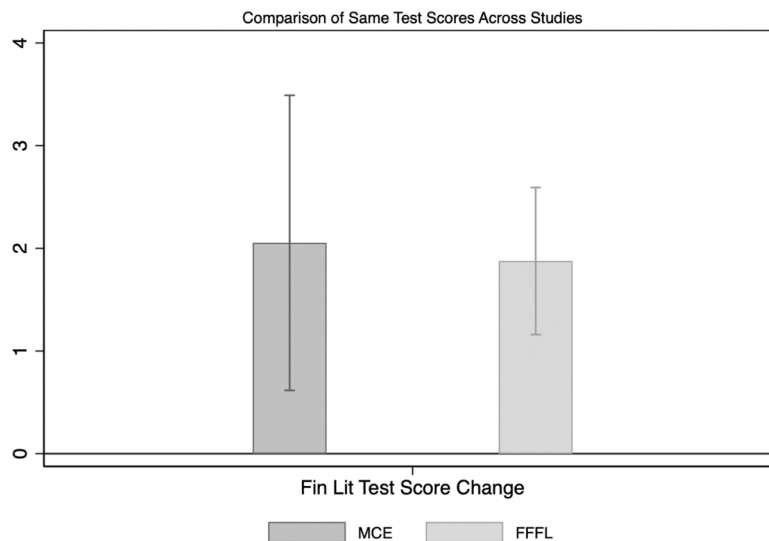


Fig. 3. Financial knowledge estimates from experiential (MCE) and traditional classroom (FFFL) programs are similar. Source: Table 7.

et al., 2015; Choi, 2009; Johnson & Sherraden, 2007). Experience making small deposits is a real-world experience that could also teach financial literacy (Choi, 2009; Collins & Odders-White, 2015). While only operating in a simulated economy, MCE offers a much simpler implementation and includes all students even if they do not have a bank account.

At its core, personal finance is an application of basic economic principles, and an activity that even elementary school students can engage in regularly. Creating opportunities for students to practice financial and economic decision making may be an effective way to let students learn from trial and error, boosting their knowledge and skills.

Providing MCE-style experiences for students from elementary grades into middle school and high school may have promise as a flexible strategy that can continue to engage students as they develop

and practice financial capability. A student who experiences a combination of experiential and classroom learning over several years might be especially likely to develop stronger financial knowledge, as well as greater financial capability in adulthood. MCE shows promise as a relatively efficient mechanism to teach financial literacy in elementary schools, and could serve as an important component of a comprehensive effort to promote financial well-being.

CRediT authorship contribution statement

Michael Batty: Conceptualization, Formal analysis. **J. Michael Collins:** Conceptualization, Formal analysis, Writing - review & editing. **Collin O'Rourke:** Data curation. **Elizabeth Odders-White:** Conceptualization, Writing - original draft.

Appendix A

A1. Teacher Fixed Effects Estimates for Financial Literacy Outcomes

Table 8

Teacher fixed effects estimates: MCE treatment vs. MCE controls for quiz scores and student reports of being taught financial literacy.

	Quiz Score	Teachers Taught
School Clustered SE		
MCE Treatment	0.845	0.539**
SE	0.501	0.244
90CI	[−0.021, 1.711]	[0.117, 0.962]
School Clustered Wild Bootstrap SE		
MCE Treatment	0.845	0.539
P	0.17	0.12
90CI	[−0.185, 1.766]	[−0.0423, 1.005]
Controls	Yes	Yes
Teacher FE	Yes	Yes
Mean	49.90	3.194
StDev	8.305	1.354
N	1633	1633
Clusters	20	20

Standard errors in parentheses * $p < .1$, ** $p < .05$, *** $p < .01$. Source: Student In-class Assessment Surveys. Teacher codes are only available for subset of students records. Teachers who worked in multiple schools in 2015 are not included.

A2. FSA Math Test Scores by Year by School Cohort

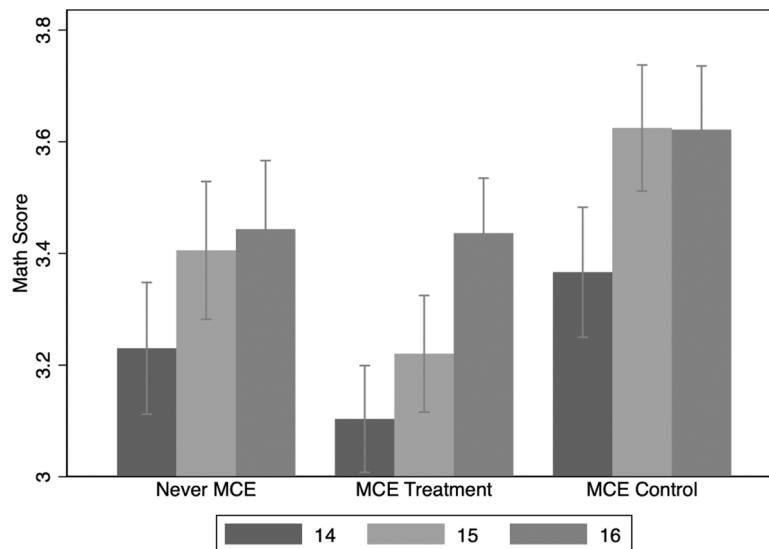


Fig. 4. Mean math achievement test score by MCE use in student's school in 2014-16. Source: SDPBC Administrative Data 2014-2016 (three school years). Conditional on students with math test score data all three years.

Table 9

Improved math test scores: additional estimates.

	(1) Up AY16 (vs 14)	(2) Up AY15 (vs 14)
Panel A: MCE Cohorts vs Never MCE: No Clustering		
MCE Treatment	0.043	−0.002
SE	(0.037)	(0.034)
90CI	[−0.029,0.115]	[−0.069,0.066]
MCE Control	−0.026	0.035
SE	(0.038)	(0.037)
90CI	[−0.101,0.050]	[−0.038,0.109]
Panel B: MCE 1st Trimester vs. MCE 2nd Trimester: Wild Bootstrap		
MCE Treatment		
Coef	0.0702	−0.0414
p	0.22	0.37
90CI	[−0.0300, 0.152]	[−0.122, 0.0365]
Clusters	20	20
Mean	0.390	0.339
StDev	0.488	0.474
N	787	802

Standard errors in parentheses * $p < .1$, ** $p < .05$, *** $p < .01$. Source: SDPBC Administrative data on Florida standardized math tests. Up AY16 (vs 14) indicates math score is higher than 2014 for that student. Up AY15 (vs 14) is an indicator of a student math score that increased over 2014.

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